

INTRINSIC DIMENSION WITH LOW RANK ADAPTORS

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BACKGROUND [1]

MOTIVATION

- We can tune a model with hundreds of millions of parameters on datasets with only hundreds or thousands of labeled examples.
- There exists a low dimension re-parameterization that is as effective as full fine-tuning.

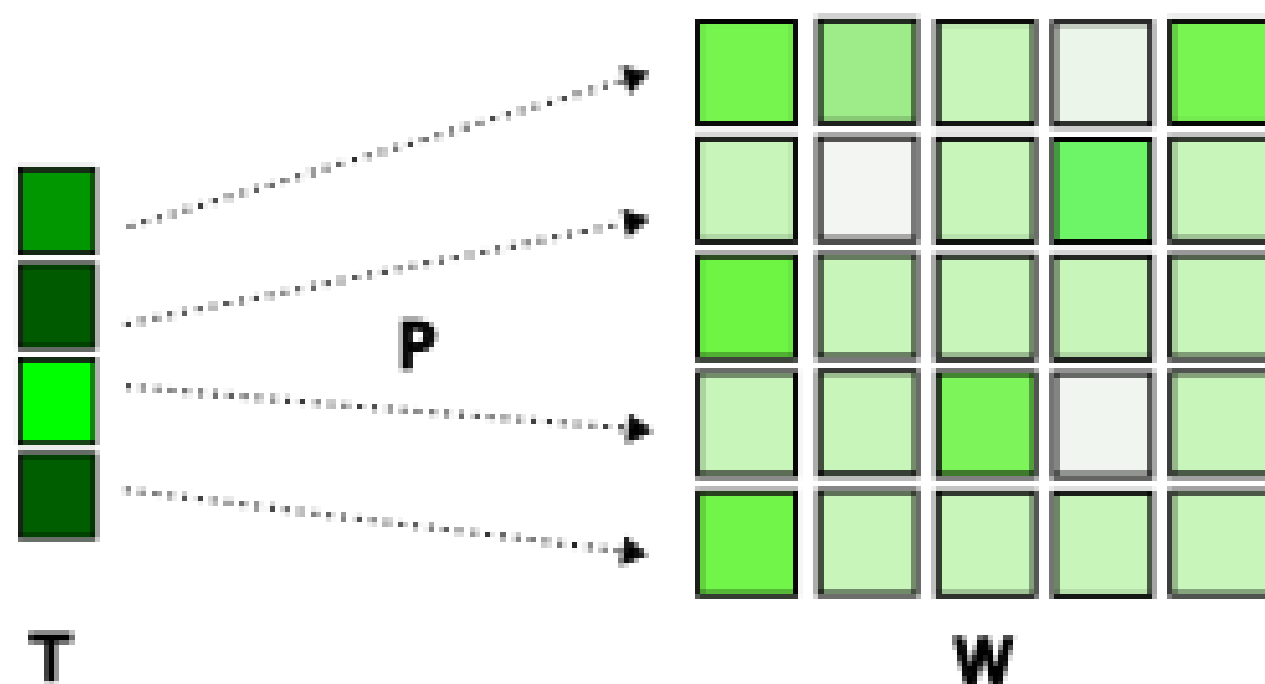
What is intrinsic dimension?

- An objective function's intrinsic dimension is defined as the minimum dimension needed to solve the problem upto some precision level.

[1] INTRINSIC DIMENSIONALITY EXPLAINS THE EFFECTIVENESS OF LANGUAGE MODEL FINE-TUNING. Aghajanyan et al

INTRINSIC DIMENSION OF FINETUNING

OVERVIEW



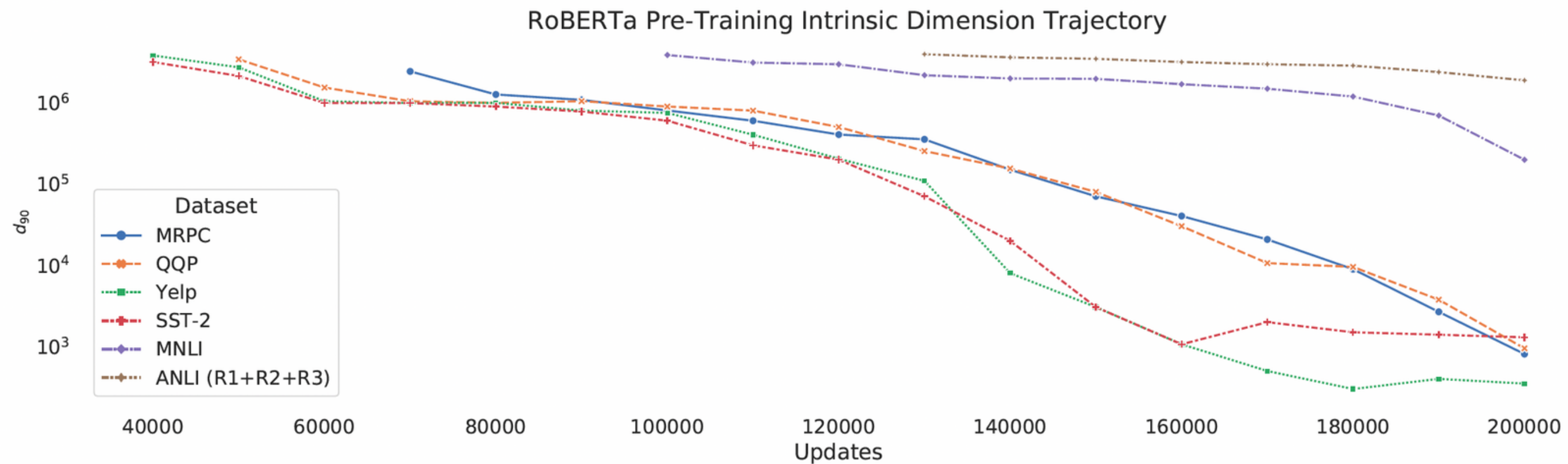
- Projection $P : T \rightarrow W$
- Intrinsic Dimension: Minimum dimension of T required under P to achieve 90% of full fine-tuning accuracy.
- Exact determination is intractable.
- Compute experimentally through search.

OBSERVATIONS

- Number of parameters is inversely correlated with intrinsic dimension. Larger models tend to have a lower intrinsic dimension.
- Intrinsic dimension can be interpreted as the minimum description length within the framework of pretrained model.

Model	SAID		DID	
	MRPC	QQP	MRPC	QQP
BERT-Base	1608	8030	1861	9295
BERT-Large	1037	1200	2493	1389
RoBERTa-Base	896	896	1000	1389
RoBERTa-Large	207	774	322	774

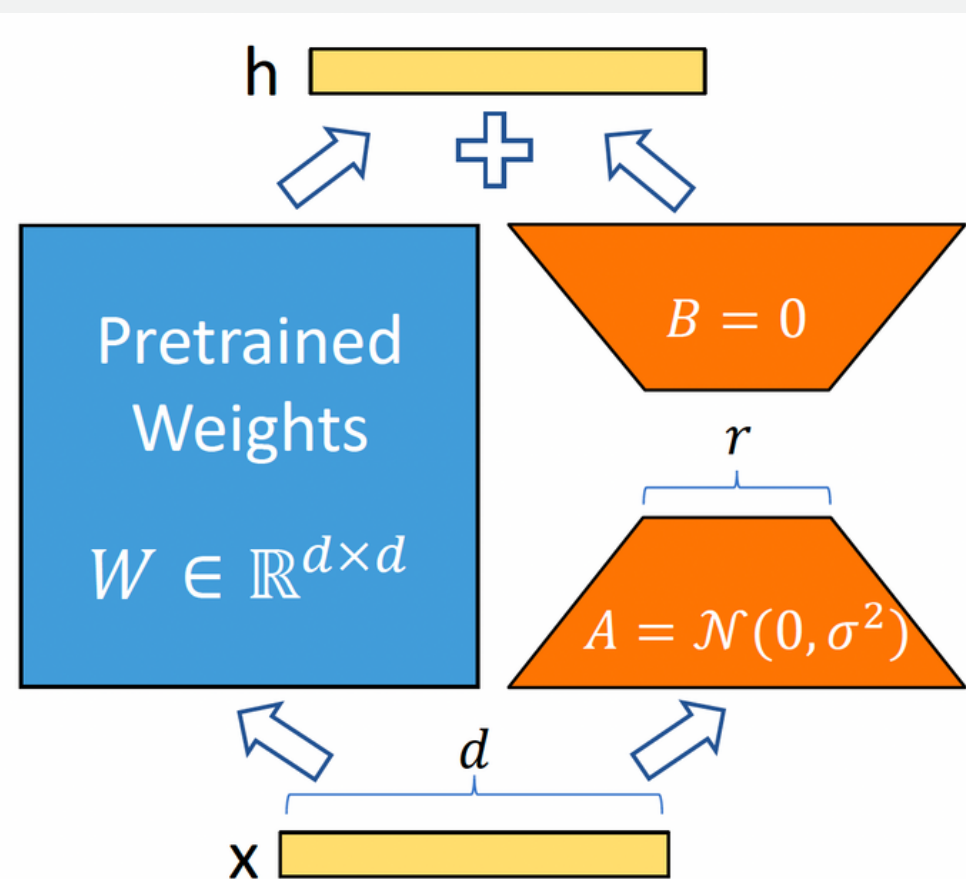
OBSERVATIONS



- Pre-training implicitly optimizes the description length of NLP tasks, without having direct access to them.

LoRA (Low Rank Adaptors) [2]

OVERVIEW



- We compute a decomposition of W with low rank matrices A & B .
- Only A , B are updated during training. W is frozen.
- Dimension of (A, B) is much lower than W .

EXPERIMENTS

PART 1: Full fine-tuning for 6 epochs

- PART 2:**
- Search through lora ranks [1, 2, 4, ... 1024] until desired accuracy is reached
 - Model is trained at each rank for 6 epochs.

Comparison across datasets and models.

RESULTS



We train BERT-Base across 4 datasets.

1. ANLI
2. MRPC
3. QQP
4. SST2

Intrinsic Dimension (d90)	LoRA Rank
<u>76,035</u>	2
38,402	1
38,402	1
38,402	1

RESULTS



We train 5 models

1. Tiny-BERT
 2. BERT-Base
 3. RoBERTa-Base
 4. BERT-Large
 5. RoBERTa-Large
- on sentiment analysis SST-2

Model Size	Intrinsic Dimension (d90)	LoRA rank
66,976,516	<u>19,970</u>	1
109,522,180	38,402	1
125,276,164	628,994	1
335,244,292	100,354	1
356,511,748	1,149,954	1



OBSERVATION

- Intrinsic dimension of larger models is larger under LoRA projection.
- Models achieve 90% accuracy at rank-1. But the approach doesn't allow to reduce parameters below rank-1.

Can we achieve higher precision with LoRAs?



RESULT



We train 5 models

1. Tiny-BERT
 2. BERT-Base
 3. RoBERTa-Base
 4. BERT-Large
 5. RoBERTa-Large
- on sentiment analysis SST-2.
Until 99% of train-accuracy.

Model Size	d (95)	d (99)
66,976,516	-	-
109,522,180	<u>38,402</u>	-
125,276,164	628,994	628,994
335,244,292	100,354	<u>100,354</u>
356,511,748	1,149,954	2,624,514



CONCLUSION

- Intrinsic dimension is related across projections. Tasks with higher intrinsic dimension under “SAID” projection has higher dimension under LoRA.
 - Most models achieve $d(90)$ at rank-1. Possible room for improvement.
 - Larger models are able to achieve higher precision at lower ranks. Smaller models struggle to achieve higher precision.
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LIMITATION

- We ran the experiments for only 6 epochs. Can we achieve better results with longer training?
 - We tested here only for training accuracy. Does intrinsic dimension for test-accuracy show similar trends?
 - Hyper-parameter tuning across learning rate, regularization and lora_alpha.
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FUTURE WORK

- More Experiments
 - Across models
 - Across datasets
 - Hyperparameter tuning and running the experiments for longer.
 - These experiments can be extended to other PEFT methods.
Other low rank methods and IA3.
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THANK YOU!

